

F2L Algorithm Guide

Sep 29, 2026 · @Archie Ellison

How to use this guide

Every case has two move sequences: the **algorithm** that solves it and a **setup** that creates it. The setup is the algorithm run backwards, so setup then algorithm on a solved cube returns you to solved. That lets you drill one case on repeat without scrambling.

How to hold the cube. White cross on the bottom, the slot you are filling at front-right. The diagrams use green front, red right, yellow top. Gray stickers are top-layer pieces that don't matter during F2L. Colored stickers are the pieces you care about.

The practice loop

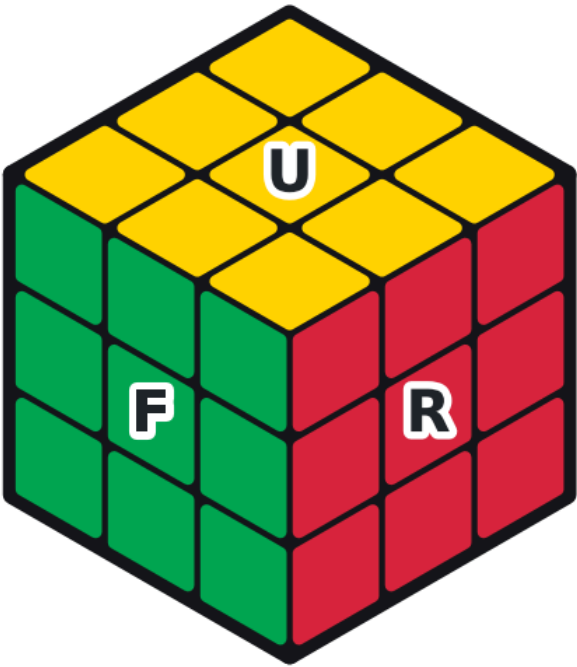
1. Start solved: white on the bottom, green in front.
2. Do the setup. Your cube now matches the diagram.
3. Do the algorithm. You are back to solved.
4. Repeat 10 times slowly with no pauses, then 10 times at speed.
5. Add recognition: do the setup, then a random U, U' or U2. Now find the case, turn the top layer until it matches the diagram, and solve.

Step 5 matters most. Drilling from solved teaches your hands the moves; it does not teach you to spot the case in a real solve. Turning the top layer to line a case up with its diagram is called the AUF (adjust U face).

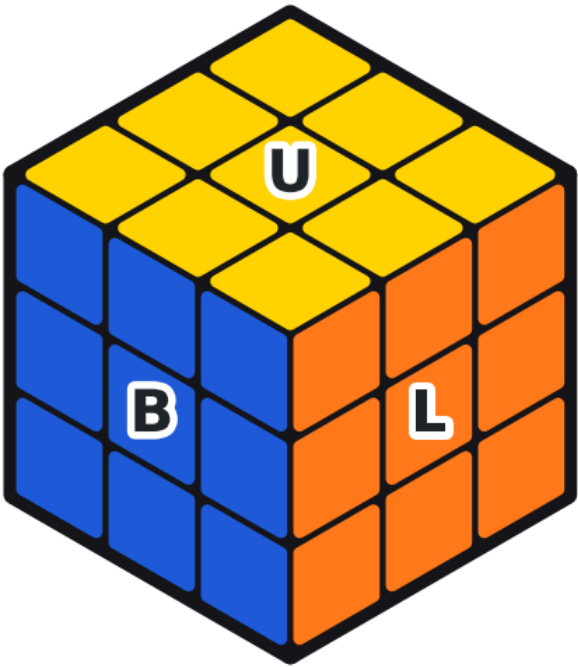
Every setup and algorithm in this guide was checked on a simulated cube: each core setup leaves the cross and the other three slots solved, and every algorithm returns all four slots to solved. A few cases from your original sheet involve two slots, so their setups disturb both.

Notation: what each letter means

A letter means: turn that face 90° clockwise, judged as if you were looking straight at that face. An apostrophe (') reverses it. A 2 means a half turn, so R2 is just R twice and the direction doesn't matter.



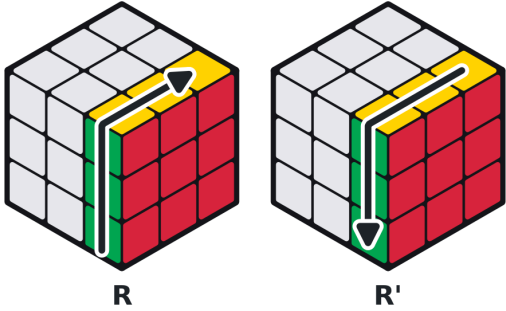
FRONT VIEW

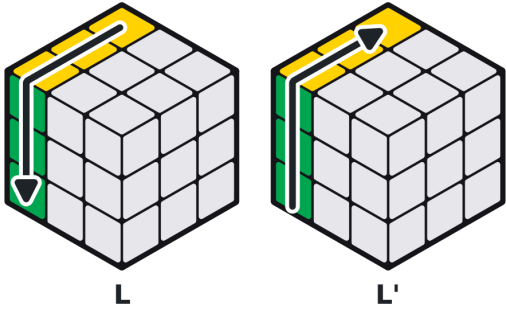
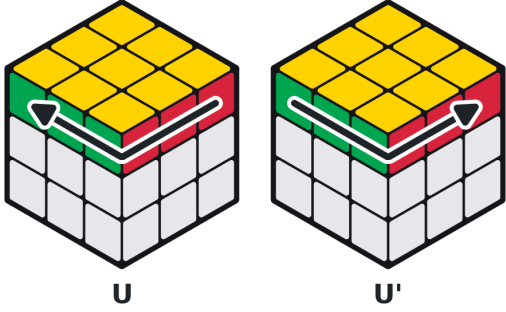
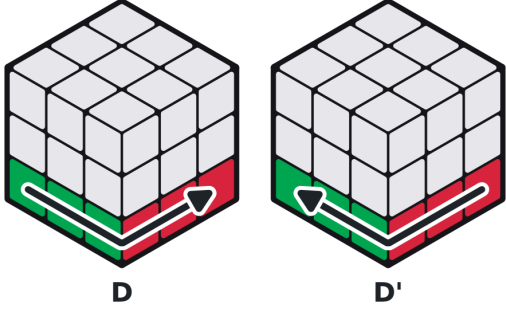
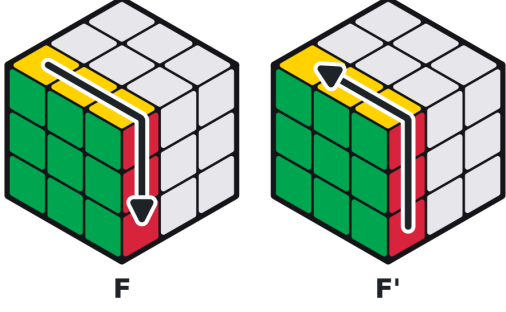


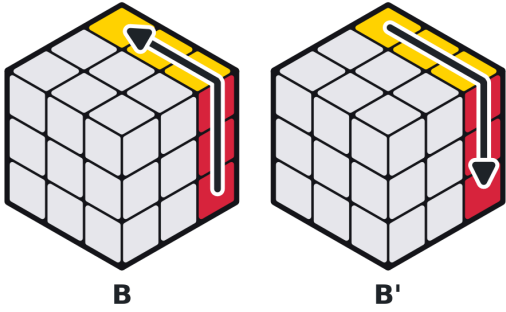
BACK VIEW

Hold white on the bottom (D) and yellow on top (U). The face pointed at you is F. In the diagrams below, colored stickers are the layer that moves, gray stickers stay put, and the arrow shows where the front stickers travel. Parentheses in an algorithm, like (R U R'), are only there to group moves so they are easier to read.

Face turns (one outer layer)

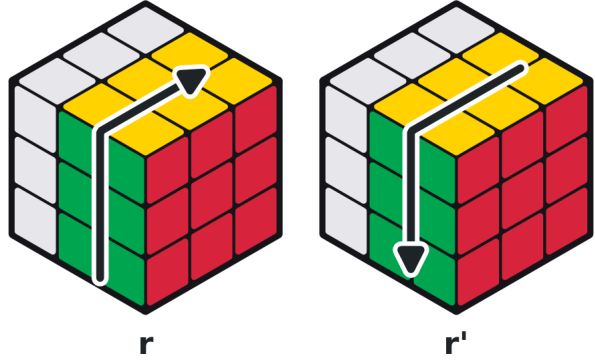
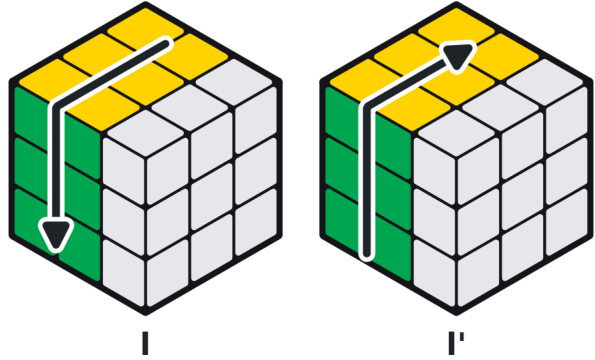
Move	Diagram	Direction
R, R'	 R R'	R: right layer, front stickers go up. R': they go down.

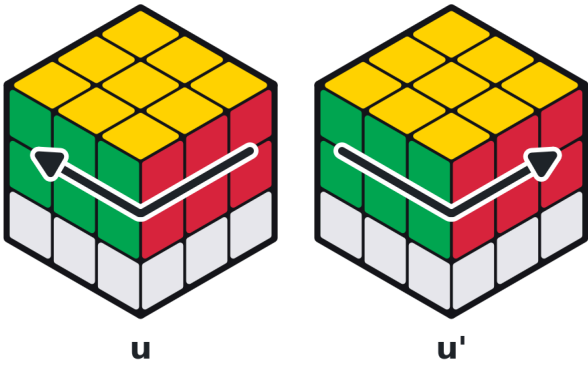
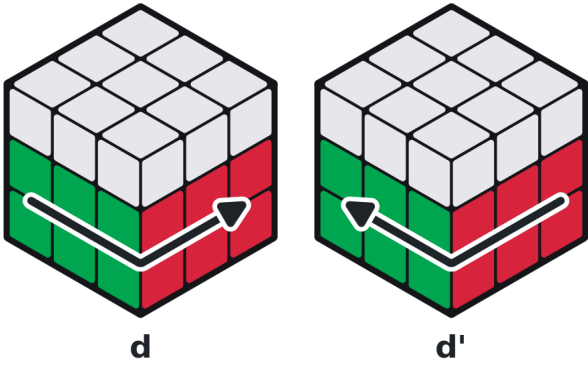
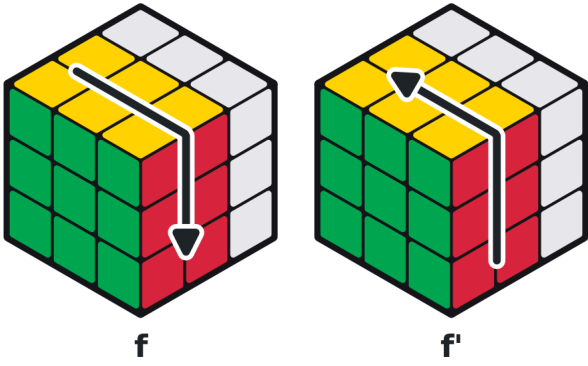
Move	Diagram	Direction
L, L'	 The diagrams for L and L' show a 3x3x3 cube with the left face highlighted in green and yellow. In the L diagram, a white arrow indicates a counter-clockwise turn of the left layer. In the L' diagram, a white arrow indicates a clockwise turn of the left layer.	L: left layer, front stickers go down. It is the opposite of R because clockwise is judged from the left side.
U, U'	 The diagrams for U and U' show a 3x3x3 cube with the top face highlighted in yellow and red. In the U diagram, a white arrow indicates a counter-clockwise turn of the top layer. In the U' diagram, a white arrow indicates a clockwise turn of the top layer.	U: top layer, front stickers go left. U': they go right.
D, D'	 The diagrams for D and D' show a 3x3x3 cube with the bottom face highlighted in green and red. In the D diagram, a white arrow indicates a counter-clockwise turn of the bottom layer. In the D' diagram, a white arrow indicates a clockwise turn of the bottom layer.	D: bottom layer, front stickers go right. The opposite of U.
F, F'	 The diagrams for F and F' show a 3x3x3 cube with the front face highlighted in green and yellow. In the F diagram, a white arrow indicates a clockwise turn of the front layer. In the F' diagram, a white arrow indicates a counter-clockwise turn of the front layer.	F: front layer turns like a clock hand, top stickers go right. F': top stickers go left.

Move	Diagram	Direction
B, B'	 The diagrams show two 3x3 cubes. In the first, labeled 'B', a white arrow on the back face (red) points left, indicating a counter-clockwise rotation. In the second, labeled 'B'', a white arrow on the back face points right, indicating a clockwise rotation. The top face is yellow and the front face is white.	B: back layer, top stickers go left. From the front it looks counterclockwise.

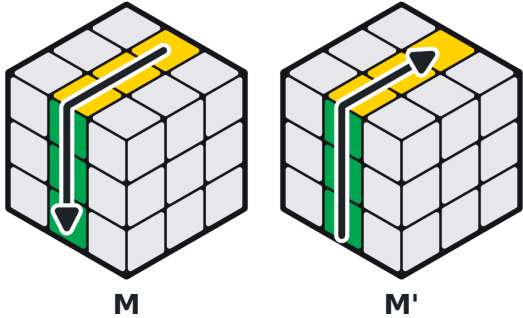
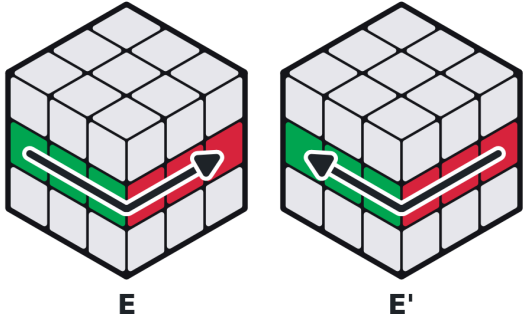
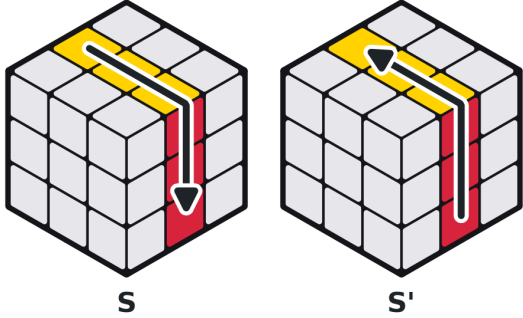
Wide turns (lowercase letter = two layers)

A lowercase letter turns the outer layer and the middle layer together. You will also see it written with a w: r and Rw are the same move.

Move	Diagram	Direction
r, r'	 The diagrams show two 3x3 cubes. In the first, labeled 'r', a white arrow on the right face (green) points right, indicating a clockwise rotation of the right two layers. In the second, labeled 'r'', a white arrow on the right face points left, indicating a counter-clockwise rotation. The top face is yellow and the front face is white.	Right two layers, same direction as R.
l, l'	 The diagrams show two 3x3 cubes. In the first, labeled 'l', a white arrow on the left face (green) points left, indicating a counter-clockwise rotation of the left two layers. In the second, labeled 'l'', a white arrow on the left face points right, indicating a clockwise rotation. The top face is yellow and the front face is white.	Left two layers, same direction as L.

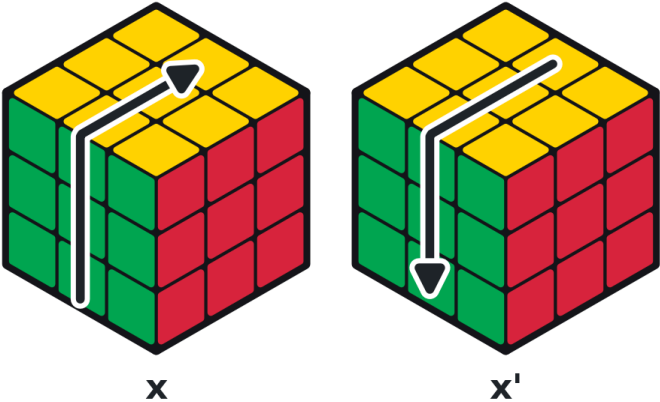
Move	Diagram	Direction
u, u'	 The diagram shows two 3x3x3 cubes. The left cube, labeled 'u', has a white arrow on the top face pointing from the center towards the front-left corner. The right cube, labeled 'u'', has a white arrow on the top face pointing from the center towards the front-right corner. Both cubes have yellow on top, green on the front, and red on the right.	Top two layers, same direction as U.
d, d'	 The diagram shows two 3x3x3 cubes. The left cube, labeled 'd', has a white arrow on the bottom face pointing from the center towards the front-right corner. The right cube, labeled 'd'', has a white arrow on the bottom face pointing from the center towards the front-left corner. Both cubes have white on top, green on the front, and red on the right.	Bottom two layers, same direction as D.
f, f'	 The diagram shows two 3x3x3 cubes. The left cube, labeled 'f', has a white arrow on the front face pointing from the top edge towards the bottom-right corner. The right cube, labeled 'f'', has a white arrow on the front face pointing from the top edge towards the bottom-left corner. Both cubes have yellow on top, green on the front, and red on the right.	Front two layers, same direction as F.

Slice turns (middle layer only)

Move	Diagram	Direction
M, M'	 M M'	Middle layer between L and R. Turns like L: front stickers go down.
E, E'	 E E'	Middle layer between U and D. Turns like D: front stickers go right.
S, S'	 S S'	Middle layer between F and B. Turns like F: top stickers go right.

Cube rotations (whole cube, nothing gets scrambled)

Rotations only change how you hold the cube. After a rotation, the letters refer to the new sides: after a y, the face that was on the right is now F.

Move	Diagram	Direction
x, x'		Whole cube turns like R.
y, y'	Image — y and y prime	Whole cube turns like U.
z, z'	Image — z and z prime	Whole cube turns like F.

If a move feels backwards, it is almost always L, D, B, M or E. Picture yourself looking straight at that face (or, for M and E, at L and D) and turn it clockwise from there.

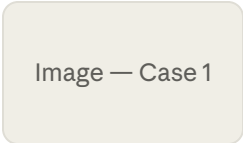
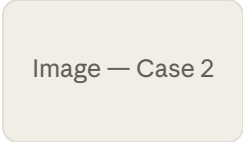
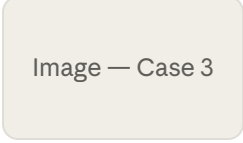
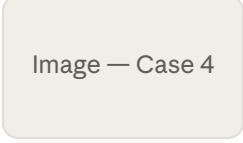
Core F2L cases: all 41

These 41 cases are every way the corner and edge for one slot can sit. Each case appears next to its mirror (the same idea done with the left hand instead of the right): #1 pairs with #2, #3 with #4, and so on. The only cases without a mirror partner are #35, #36 and #37. Numbers are this guide's own, not the numbering on other sites.

The algorithm column is a fast, widely used solution. The setup column is what you do on a solved cube to create the case. The diagram shows the cube right before you start the algorithm, including any U turn at the start.

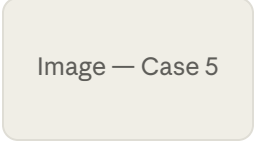
Basic inserts (1 to 4)

Every other case ends in one of these four, so make them automatic first. In #1 and #2 the edge waits on the far side and one trigger joins and inserts the pair. In #3 and #4 the pair is already built: move it into place and insert.

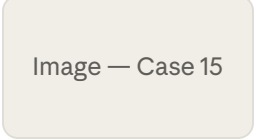
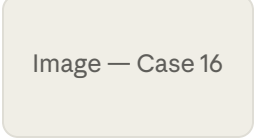
#	Case	Algorithm	Setup	Look for
1	 Image — Case 1	$R\ U\ R'$	$R\ U'\ R'$	White right; edge back, green on top
2	 Image — Case 2	$F'\ U'\ F$	$F'\ U\ F$	White front; edge left, red on top
3	 Image — Case 3	$U\ R\ U'$ R'	$R\ U\ R'$ U'	White front; edge right, green on top
4	 Image — Case 4	$U'\ F'\ U$ F	$F'\ U'\ F$ U	White right; edge front, red on top

Corner on top, white facing the side (5 to 16)

Both pieces are on top but not joined correctly. Each algorithm does the same three jobs: move the edge so a pair can form, join it to the corner, then finish with a basic insert. Recognize these by where the edge sits and which color is on its top.

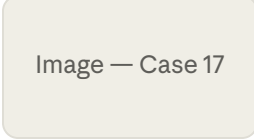
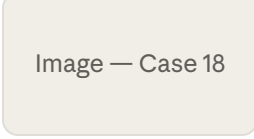
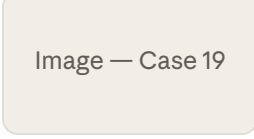
#	Case	Algorithm	Setup	Look for
5	 Image — Case 5	$U'\ R\ U\ R'\ U\ R\ U$ R'	$R\ U'\ R'\ U'\ R$ $U'\ R'\ U$	White right; edge left, green on top

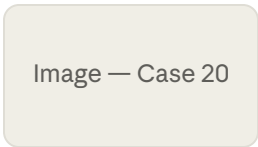
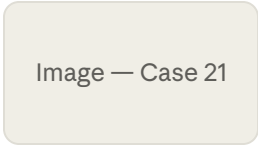
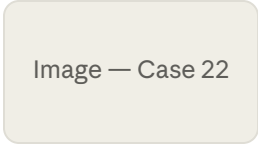
#	Case	Algorithm	Setup	Look for
6	Image — Case 6	U F' U' F U' F' U' F	F' U F U F' U F U'	White front; edge back, red on top
7	Image — Case 7	U' R U' R' U R U R'	R U' R' U' R U R' U	White right; edge right, green on top
8	Image — Case 8	U F' U F U' F' U' F	F' U F U F' U' F U'	White front; edge front, red on top
9	Image — Case 9	U' R U2 R' U2 R U' R'	R U R' U2 R U2 R' U	White front; edge left, green on top
10	Image — Case 10	U F' U2 F U2 F' U F	F' U' F U2 F' U2 F U'	White right; edge back, red on top
11	Image — Case 11	U' R U R' U2 R U' R'	R U R' U2 R U' R' U	White front; edge back, green on top
12	Image — Case 12	U' r U' R' U R U r'	r U' R' U' R U r' U	White right; edge left, red on top
13	Image — Case 13	U' R U2 R' U F' U' F	F' U F U' R U2 R' U	White front; edge right, red on top

#	Case	Algorithm	Setup	Look for
14		$R' U2 R2 U R2 U$ R	$R' U' R2 U' R2$ $U2 R$	White right; edge front, green on top
15		$R U' R' U2 F'$ $U' F$	$F' U F U2 R U$ R'	White right; edge right, red on top
16		$F' U F U2 R U$ R'	$R U' R' U2 F'$ $U' F$	White front; edge front, green on top

Corner on top, white facing up (17 to 24)

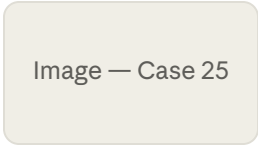
White on top is the hardest thing to pair by intuition, and where algorithms save the most moves. The standard fix is $R U2 R'$ or $F' U2 F$ (#17 to #20): it swings the corner around so white faces the side, which turns the case into an easy one.

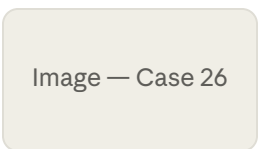
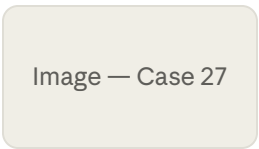
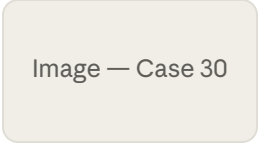
#	Case	Algorithm	Setup	Look for
17		$R U2 R' U' R U$ R'	$R U' R' U R U2$ R'	White up; edge right, green on top
18		$F' U2 F U F' U'$ F	$F' U F U' F'$ $U2 F$	White up; edge front, red on top
19		$U R U2 R' U R U'$ R'	$R U R' U' R U2$ $R' U'$	White up; edge back, green on top

#	Case	Algorithm	Setup	Look for
20		$U' F' U2 F U' F'$ $U F$	$F' U' F U F'$ $U2 F U$	White up; edge left, red on top
21		$U2 R U R' U R U'$ R'	$R U R' U' R U'$ $R' U2$	White up; edge left, green on top
22		$r U' r' U2 r U$ r'	$r U' r' U2 r U$ r'	White up; edge back, red on top
23		$U2 R2 U2 R' U' R$ $U' R2$	$R2 U R' U R U2$ $R2 U2$	White up; edge front, green on top
24		$F U R U' R' F' R$ $U' R'$	$R U R' F R U$ $R' U' F'$	White up; edge right, red on top

Corner in the slot, edge on top (25 to 30)

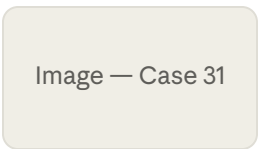
Line the edge up where the diagram shows it, then the first moves lift the corner out already joined to its edge, and the rest puts the pair back in correctly. None of these are longer than 8 moves.

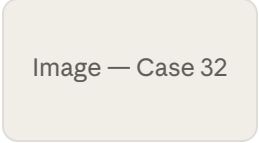
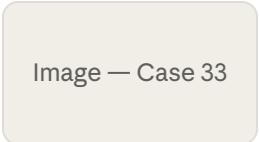
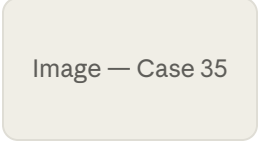
#	Case	Algorithm	Setup	Look for
25		$U R U' R' U'$ $F' U F$	$F' U' F U R$ $U R' U'$	Corner solved in slot; edge front, red on top

#	Case	Algorithm	Setup	Look for
26		U' F' U F U R U' R'	R U R' U' F' U' F U	Corner solved in slot; edge right, green on top
27		F' U F U' F' U F	F' U' F U F' U' F	Corner in slot, white right; edge front, red on top
28		R U' R' U R U' R'	R U R' U' R U R'	Corner in slot, white front; edge right, green on top
29		R U R' U' R U R'	R U' R' U R U' R'	Corner in slot, white right; edge right, green on top
30		F' U' F U F' U' F	F' U F U' F' U F	Corner in slot, white front; edge front, red on top

Edge in the slot, corner on top (31 to 36)

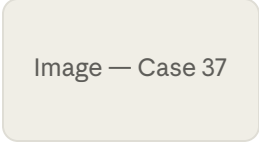
The edge is stuck in the slot. Each algorithm pulls the edge out paired with the corner, or sets up a basic insert. #36 is R U R' U' twice, then R U R'.

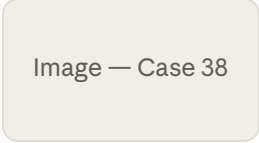
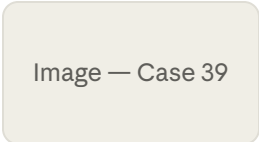
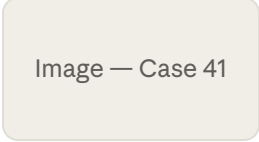
#	Case	Algorithm	Setup	Look for
31		U R U R' U2 R U R'	R U' R' U2 R U' R' U'	White right; edge in slot, correct

#	Case	Algorithm	Setup	Look for
32		U' R U' R' U2 R U' R'	R U R' U2 R U R' U	White front; edge in slot, correct
33		U2 F' U F U R U R'	R U' R' U' F' U' F U2	White right; edge in slot, flipped
34		U2 R U' R' U' F' U' F	F' U F U R U R' U2	White front; edge in slot, flipped
35		U' R' F R F' R U' R'	R U R' F R' F' R U	White up; edge in slot, flipped
36		R U R' U' R U R' U' R U R'	R U' R' U R U' R' U R U' R'	White up; edge in slot, correct

Both pieces in the slot, solved wrong (37 to 41)

The pieces are in the right slot but twisted or flipped. Take the pair out as a unit and put it back in correctly. These are rare in real solves, so learn them last.

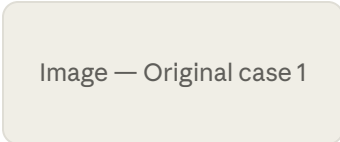
#	Case	Algorithm	Setup	Look for
37		R2 U2 F R2 F' U2 R' U R'	R U' R U2 F R2 F' U2 R2	Corner solved in slot; edge in slot, flipped

#	Case	Algorithm	Setup	Look for
38		$R\ U'\ R'\ U'\ R\ U$ $R'\ U^2\ R\ U'\ R'$	$R\ U\ R'\ U^2\ R$ $U'\ R'\ U\ R\ U$ R'	Corner in slot, white front; edge in slot, correct
39		$R\ U'\ R'\ U\ R\ U^2$ $R'\ U\ R\ U'\ R'$	$R\ U\ R'\ U'\ R$ $U^2\ R'\ U'\ R\ U$ R'	Corner in slot, white right; edge in slot, correct
40		$R\ F\ U\ R\ U'\ R'\ F'$ $U'\ R'$	$R\ U\ F\ R\ U\ R'$ $U'\ F'\ R'$	Corner in slot, white front; edge in slot, flipped
41		$R\ U'\ R'\ r\ U'\ r'$ $U^2\ r\ U\ r'$	$r\ U'\ r'\ U^2\ r$ $U\ r'\ R\ U\ R'$	Corner in slot, white right; edge in slot, flipped

Your original sheet, recreated

These are the 10 algorithms from [F2L Cases \(#cuber\)](#), redrawn and checked. Five solve cases already in the core set (S8 is a move faster than the core version, S9 is slower). One solves two slots at once. Four fix an edge stuck in the wrong slot. The bracketed reverses on the original sheet are all correct: they are the setups below.

The original photographed most cases with red in front. The diagrams here show each case held the normal way (white bottom, green front), which does not change the algorithm because every move is relative to how you hold the cube. When a piece is in a back slot, a second view shows the back.

#	Case	Algorithm	Setup	What it is
S1		$r\ U'\ R'\ U\ R$ $U\ r'$	$r\ U'\ R'\ U'$ $R\ U\ r'$	Same case as #12, one U' later. Use whichever starting position you spot first.

#	Case	Algorithm	Setup	What it is
S2	Image — Original case 2	$L F' L' F R$ $U R'$	$R U' R' F'$ $L F L'$	Two slots at once. With front-left also open, $L F' L' F$ solves the front-left pair and sets up the front-right pair, then $R U R'$ inserts it. The front-right pieces alone are case #23.
S3	Image — Original case 3	$r U' r' U2 r$ $U r'$	$r U' r' U2$ $r U r'$	Identical to #22.
S4	Image — Original case 4	$R2 U2 R' U'$ $R U' R2$	$R2 U R' U$ $R U2 R2$	Same case as #23, with the corner at back-left so you skip the opening $U2$.
S5	Image — Original case 5	$R2 u R2 u'$ $R2$	$R2 u R2 u'$ $R2$	Edge swap. White-up corner above the slot, its edge stuck in the back-left slot, and the back-left edge sitting in your front-right slot. Fixes both slots.
S6	Image — Original case 6	$R2 u' R2 u$ $R2$	$R2 u' R2 u$ $R2$	Edge swap on other slots. Back-right corner on top with white up, back-right edge stuck in the front-left slot, front-left edge in the back-right slot. Fixes both.

#	Case	Algorithm	Setup	What it is
S7	Image — Original case 7	L2 u L2 u' L2	L2 u L2 u' L2	Front-right corner already solved, but its edge is in the back-left slot and the back-left edge is in the front-right slot. Back-left corner waits on top, white up. Fixes both.
S8	Image — Original case 8	r' U2 R2 U R2 U r	r' U' R2 U' R2 U2 r	Same case as #10. One move shorter and no F turns; try both and keep the faster one.
S9	Image — Original case 9	r' D' r U' r' D r	r' D' r U r' D r	Same case as #4, a basic insert. U2 F' U F solves this exact position in 4 moves, so this one is not worth learning.
S10	Image — Original case 10	R u R' U R u' R'	R u R' U' R u' R'	Front-right corner on top at back-left, white up. Its edge is stuck in the back-left slot, and the back-left edge is on top at the left. Fixes both.

The original sheet also links this [video walkthrough](#).

Practice plan

Don't learn all 41 at once. Your intuitive F2L already handles the easy cases well, so start where intuition costs you the most moves: white facing up and pieces stuck in the slot. Only keep an algorithm if it beats what you would do intuitively.

Learning order (about one block per week at 15 minutes a day)

- ☐ Block 1: #17 to #24, corner white facing up
- ☐ Block 2: #31 to #36, edge stuck in the slot
- ☐ Block 3: #25 to #30, corner stuck in the slot

- ☐ Block 4: #5 to #16, white facing the side (keep only the ones faster than your intuitive solution)
- ☐ Block 5: #37 to #41, plus S5, S6, S7 and S10 from the original sheet

Daily 15-minute drill

1. Pick two mirror pairs (4 cases) from the current block.
2. For each case: setup then algorithm, 10 times slowly with no pauses, then 10 times at speed.
3. Recognition round: setup, random U turn, then find the case and solve it. Repeat until you get each one right 5 times in a row without thinking.
4. Finish with 5 full solves. Each time a case from your current block shows up, use the algorithm even if it feels slower at first.

A case is learned when you can spot it from any U position and start turning within a second. Turning slowly without pauses beats turning fast and stopping to look.